

Code Explanation

E2E Address Change YUYU Processing Code Explanation

Overview

The provided code is the main module for E2E Address Change YUYU processing. It is a PySpark job that processes address change records for YUYU output. The job takes command line arguments, initializes a Spark session, and uses an `AddressChangeProcessor` class to process the data.

Step 1: Parsing Command Line Arguments

The code starts by parsing command line arguments using the `argparse` library. This step is necessary to provide the job with the required configuration and input parameters.

```
def parse_arguments():
    parser = argparse.ArgumentParser(description='Process address change records for YUYU output')
    parser.add_argument('--process-userid', required=True, help='User ID for process stamping')
    parser.add_argument('--source-table', default='ZSYSE2EDEV.STG_E2E_AC_TXDBH_DATA',
                        help='Source staging table name')
    parser.add_argument('--target-table', default='ZSYSE2EDEV.STG_E2E_AC_TXDBH_DATA',
                        help='Target staging table name')
    parser.add_argument('--output-path', default='/tmp/yuyu_output/DAddressChangeYUYU',
                        help='Output path for YUYU extract')
    return parser.parse_args()
```

Step 2: Initializing the Main Entry Point

The `main` function is the entry point for the job. It calls the `parse\_arguments` function to parse the command line arguments.

```
def main():
    args = parse_arguments()
```

Step 3: Initializing Spark Session

The code initializes a Spark session with Hive support using the `SparkSession.builder` API. This step is necessary to interact with Hive tables and perform data processing.

```
spark = SparkSession.builder
    .appName("E2E Address Change YUYU Processing")
    .enableHiveSupport()
    .getOrCreate()
```

Step 4: Initializing Address Change Processor

The code initializes an instance of the `AddressChangeProcessor` class, passing the Spark session and the `process\_userid` argument to its constructor.

```
processor = AddressChangeProcessor(spark, args.process_userid)
```

Step 5: Executing Address Change Processing

The code calls the `process` method of the `AddressChangeProcessor` instance, passing the `source_table`, `target_table`, and `output_path` arguments. This step performs the actual data processing.

```
processor.process(args.source_table, args.target_table, args.output_path)
```

Step 6: Stopping Spark Session

Finally, the code stops the Spark session to release resources.

```
spark.stop()
```

Step 7: Guarding the Main Entry Point

The code uses the `if __name__ == "__main__":` guard to ensure that the `main` function is only executed when the script is run directly, not when it is imported as a module.

```
if __name__ == "__main__":  
    main()
```